

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2(d\theta^2 + \sin^2\theta d\varphi^2)$$

$$\rightarrow G = c = M = 1$$

① Condition de normalisation

$\rightarrow$ On pose τ le temps propre :

$$ds^2 = -d\tau^2 \quad \text{On divise par } d\tau^2$$

$$\frac{ds^2}{d\tau^2} = -1 = g_{\mu\nu} u^\mu u^\nu$$

Dans le plan équatorial : $\theta = \frac{\pi}{2}$, $\frac{d\theta}{d\tau} = 0$

$$\Rightarrow 2\mathcal{L} = -\left(1 - \frac{2}{r}\right) \left(\frac{dt}{d\tau}\right)^2 + \left(1 - \frac{2}{r}\right)^{-1} \left(\frac{dr}{d\tau}\right)^2 + r^2 \left(\frac{d\varphi}{d\tau}\right)^2 = -1$$

② Constantes du mouvement

Euler-Lagrange : $\frac{d}{d\tau} \left(\frac{\partial \mathcal{L}}{\partial \left(\frac{dq}{d\tau}\right)} \right) - \frac{\partial \mathcal{L}}{\partial q} = 0$ où q est une coordonnée quelconque

Pour t $\frac{\partial \mathcal{L}}{\partial t} = 0$

$$\Rightarrow \frac{d}{d\tau} \left(\frac{\partial \mathcal{L}}{\partial \left(\frac{dt}{d\tau}\right)} \right) = 0$$

On dérive $2\mathcal{L}$ par rapport à $\frac{dt}{d\tau}$:

$$2\mathcal{L} = -\left(1 - \frac{2}{r}\right) \left(\frac{dt}{d\tau}\right)^2 + \text{termes sans } \frac{dt}{d\tau}$$

$$\frac{\partial(2\mathcal{L})}{\partial \left(\frac{dt}{d\tau}\right)} = -2\left(1 - \frac{2}{r}\right) \frac{dt}{d\tau}$$

Donc $\frac{\partial \mathcal{L}}{\partial \left(\frac{dt}{d\tau}\right)} = -\left(1 - \frac{2}{r}\right) \frac{dt}{d\tau} = -E$

$$\Rightarrow \boxed{E = \left(1 - \frac{2}{r}\right) \frac{dt}{d\tau}} \Rightarrow \frac{dt}{d\tau} = E \left(1 - \frac{2}{r}\right)^{-1}$$

Pour φ $\frac{\partial \mathcal{L}}{\partial \varphi} = 0$ et $\frac{\partial \mathcal{L}}{\partial \left(\frac{d\varphi}{d\tau}\right)} = \text{cste}$

$$\frac{\partial(2\mathcal{L})}{\partial \left(\frac{d\varphi}{d\tau}\right)} = 2r^2 \frac{d\varphi}{d\tau} = \text{cste} = 2L$$

$$\Rightarrow \boxed{L = r^2 \frac{d\varphi}{d\tau}} \Rightarrow \frac{d\varphi}{d\tau} = \frac{L}{r^2}$$

③ Equation d'énergie et potentiel effectif

On substitue dans la condition de normalisation

$$-(1 - \frac{2}{r})(\frac{E}{1 - \frac{2}{r}})^2 + (1 - \frac{2}{r})^{-1} (\frac{dr}{d\tau})^2 + r^2 (\frac{L}{r^2})^2 = -1$$

$$\text{Or } -(1 - \frac{2}{r})(\frac{E}{1 - \frac{2}{r}})^2 = -(1 - \frac{2}{r}) \cdot (\frac{E^2}{(1 - \frac{2}{r})^2}) = -\frac{E^2}{1 - \frac{2}{r}}$$

L'équation devient:

$$-\frac{E^2}{1 - \frac{2}{r}} + (1 - \frac{2}{r})^{-1} (\frac{dr}{d\tau})^2 + \frac{L^2}{r^2} = -1$$

On multiplie par $1 - \frac{2}{r}$

$$-E^2 + (\frac{dr}{d\tau})^2 + \frac{L^2}{r^2} (1 - \frac{2}{r}) = -(1 - \frac{2}{r})$$

$$-E^2 + (\frac{dr}{d\tau})^2 + \frac{L^2}{r^2} - \frac{2L^2}{r^3} = -1 + \frac{2}{r}$$

$$(\frac{dr}{d\tau})^2 = E^2 - 1 + \frac{2}{r} - \frac{L^2}{r^2} + \frac{2L^2}{r^3}$$

On divise par 2

$$\frac{1}{2} (\frac{dr}{d\tau})^2 = \frac{E^2 - 1}{2} + \frac{1}{r} - \frac{L^2}{2r^2} + \frac{L^2}{r^3}$$

④ ~~Accélération radiale~~ $\frac{d^2 r}{d\tau^2}$

$$\frac{1}{2} (\frac{dr}{d\tau})^2 = \frac{1}{r} + \frac{L^2}{2r^2} - \frac{L^2}{r^3} = \frac{E^2 - 1}{2}$$

On pose $V_{\text{eff}}(r) = -\frac{1}{r} + \frac{L^2}{2r^2} - \frac{L^2}{r^3}$ le potentiel effectif

$$\text{Ainsi : } \frac{1}{2} (\frac{dr}{d\tau})^2 + V_{\text{eff}}(r) = \frac{E^2 - 1}{2} \quad (1)$$

⑤ Accélération radiale $\frac{d^2 r}{d\tau^2}$

On dérive (1) par rapport à τ

$$\frac{dr}{d\tau} \cdot \frac{d^2 r}{d\tau^2} + \frac{dV_{\text{eff}}(r)}{dr} \cdot \frac{dr}{d\tau} = 0$$

On divise par $\frac{dr}{d\tau}$:

$$\frac{d^2 r}{d\tau^2} = -\frac{dV_{\text{eff}}(r)}{dr}$$

Calcul terme par terme de $\frac{dV_{\text{eff}}(r)}{dr}$:

$$\frac{d}{dr}(-\frac{1}{r}) = \frac{1}{r^2}; \quad \frac{d}{dr}(\frac{L^2}{2r^2}) = -\frac{L^2}{r^3}; \quad \frac{d}{dr}(-\frac{L^2}{r^3}) = \frac{3L^2}{r^4}$$

Ainsi, on obtient:

$$\boxed{\frac{d^2 r}{d\tau^2} = -\frac{1}{r^2} + \frac{L^2}{r^3} - \frac{3L^2}{r^4}}$$